

Technical Documentation

LumiNex SaaS Platform — Architecture, APIs, Structure & Enhancement Specs

1. Technology Stack

Category	Technology	Version	Purpose
Framework	Angular	21.2.x	SPA frontend with lazy-loaded modules
UI Library	Angular Material	21.2.10	Pre-built accessible components
CSS Framework	Bootstrap	5.3.8	Responsive grid & utility classes
Icons	Bootstrap Icons + FontAwesome	1.13 / 7.x	Visual iconography throughout UI
HTTP	Angular HttpClient	Built-in	REST API communication
Reactive	RxJS	7.8.x	Observable streams & async state
Auth	JWT + @auth0/angular-jwt	5.2.0	Token decode & HTTP auto-attach
Backend (dev)	json-server	1.0.0-beta	Mock REST API on port 3000
Backend (prod)	Express.js + Node.js	5.2.1	Production API server
Language	TypeScript	5.9.x	Strict typing across codebase
Build Tool	Vite (via Angular CLI)	21.2.9	Fast HMR & production bundling
Deployment	Vercel (vercel.json configured)	—	CDN hosting + CI/CD pipeline
Package Mgr	npm	11.12.1	Dependency management
ID Generation	uuid / nanoid	14.x	Unique IDs for records

2. Project Structure

All source code lives under **src/app/**. The project follows Angular's recommended lazy-loading pattern with feature-based folder organization and a shared core for guards, interceptors, models, and services.

src/app/admin/	Admin role — all admin-facing views
all-requests/	View, filter, and update all client service requests
client-management/	CRUD interface for managing registered clients
dashboard/	KPI cards — revenue, requests, team performance
employee-management/	Add, view, and remove employee accounts
payments/	Payment list, status badges, and admin controls
revenue/	Revenue charts, totals, and payment analytics
service-management/	CRUD for services, categories, and required documents
src/app/client/	Client role — self-service portal
catalogue/	Browse all active services with category filter
dashboard/	Personal overview — active requests and plan
my-requests/	List all submitted service requests with status
payments/	Payment submission form + payment history
plans/	Subscription plan selector and upgrade flow
profile/	Edit personal and company information
request-detail/	Timeline view for a single request's lifecycle
request-form/	Multi-field form to submit a new service request
subscriptions/	Active subscription status and renewal info
src/app/employee/	Employee role — task management panel
my-tasks/	List of tasks assigned by admin
summary/	Personal stats — completed, in-progress, pending
task-detail/	Update progress %, add notes, view client info
src/app/pages/	Public pages — no authentication required
home/	Landing page with hero, services, and CTA
contact/	Contact form with Bangladesh location picker
packages/	Public pricing page with subscription plan cards

<code>src/app/auth/</code>	Authentication flows
<code>login/</code>	Login form with JWT generation and role redirect
<code>register/</code>	Multi-step client registration with location fields
<code>src/app/core/</code>	Shared infrastructure (no UI)
<code>guards/</code>	<code>auth.guard</code> / <code>role-guard</code> / <code>login.guard</code>
<code>interceptors/</code>	<code>auth.interceptor</code> (JWT header) / <code>loading.interceptor</code>
<code>models/</code>	TypeScript interfaces: User, Service, ServiceRequest, Payment, Notification
<code>services/</code>	API service classes for each data entity
<code>src/app/shared/components/</code>	Reusable UI components used across all roles
<code>navbar/</code>	Top navigation bar with role-aware links
<code>sidebar/</code>	Role-specific side navigation with route links
<code>footer/</code>	Page footer with branding and links
<code>notification-bell/</code>	Bell icon with unread count and dropdown
<code>loading-bar/</code>	Global HTTP loading progress bar
<code>request-timeline/</code>	Visual status steps for a request's progress
<code>status-badge/</code>	Color-coded pill badge for request/payment status

3. Auth Flow & Route Guards

Authentication uses **JWT tokens stored in localStorage**. Three guards protect all routes. The **auth.interceptor** automatically attaches the Authorization header to every HTTP call. The **loading.interceptor** manages the global loading bar.

Guard File	Protects	Logic
auth.guard.ts	All /admin, /client, /employee routes	Checks localStorage for JWT. Redirects to /login if missing.
role-guard.ts	Role-specific routes	Reads user.role from JWT payload. Checks against route data.allowedRoles[]. Redirects unauthorized.
login.guard.ts	/login and /register	If user already has valid JWT, redirects to their role dashboard. Prevents re-login.

Login Flow

Step	Action	Result
1	User submits email + password	POST /users?email=&password;= to json-server
2	Response received	User object with role returned
3	JWT created (mock)	Token stored in localStorage as 'luminex_token'
4	User object stored	Stored in localStorage as 'luminex_user'
5	Role-based redirect	ADMIN /admin CLIENT /client EMPLOYEE /employee

4. REST API Endpoints (json-server, Port 3000)

json-server exposes standard RESTful endpoints from **db.json**. In production, these are replaced by an Express.js server with proper auth middleware.

Method	Endpoint	Description	Access
GET	/users	List all users	Admin
GET	/users/:id	Get single user by ID	Any (own)
POST	/users	Register a new user	Public
PATCH	/users/:id	Update user profile fields	Owner / Admin
DELETE	/users/:id	Remove user account	Admin
GET	/categories	List service categories	Public
POST	/categories	Create new category	Admin
PATCH	/categories/:id	Update category name/icon/color	Admin
GET	/services	List all services (with filters)	Public
GET	/services?isActive=true	List only active services	Public
POST	/services	Create new service	Admin
PATCH	/services/:id	Update service details / toggle active	Admin
DELETE	/services/:id	Delete service	Admin
GET	/service-requests	List all requests	Admin
GET	/service-requests?userId=:id	List requests for one client	Client
GET	/service-requests?assignedTo=:id	List tasks for one employee	Employee
POST	/service-requests	Submit a new service request	Client
PATCH	/service-requests/:id	Update status, progress, notes	Admin / Employee
GET	/payments	List all payments	Admin
GET	/payments?clientId=:id	Payments for one client	Client

Method	Endpoint	Description	Access
POST	/payments	Record a new payment	Client / Admin
PATCH	/payments/:id	Update payment status	Admin
GET	/subscriptions	List all subscription plans	Public
POST	/subscriptions	Create new plan (proposed)	Admin
PATCH	/subscriptions/:id	Edit plan price/features (proposed)	Admin
GET	/notifications?userId=:id	Get notifications for a user	Any (own)
PATCH	/notifications/:id	Mark notification as read	Any (own)
POST	/notifications	Create a notification	System

5. Proposed New API Endpoints (After Enhancements)

Method	Endpoint	Description	Feature
GET	/theme-settings/1	Load global UI theme (colors, logo, font)	Dynamic UI
PATCH	/theme-settings/1	Admin saves new theme settings	Dynamic UI
GET	/site-content	Load editable home page content	Content Editor
PATCH	/site-content/:id	Admin updates hero/about/contact text	Content Editor
POST	/invoices	Generate and store invoice record	Invoices
GET	/invoices?clientId=:id	Client's invoice list	Invoices
POST	/chat-messages	Send a chat message	Live Chat
GET	/chat-messages?requestId=:id	Load chat for a request	Live Chat
POST	/uploads	Upload file (multipart/form-data)	File Upload
GET	/reviews?requestId=:id	Get client review for completed request	Reviews
POST	/reviews	Submit star rating + comment	Reviews
POST	/payment/initiate	Start SSLCommerz payment session	Real Payments
POST	/payment/verify	Webhook: verify SSLCommerz callback	Real Payments

6. Full Enhanced Project Structure (After All Additions)

The table below shows every new component, service, and module to add. Existing items are marked (existing). New items are marked (NEW).

Path	Status	Description
admin/theme-builder/	NEW	Admin UI to pick brand color, accent, logo URL, font family — saves to /theme-settings
admin/package-builder/	NEW	Reactive form to create/edit subscription plans with drag-reorder feature list
admin/invoice-management/	NEW	View all invoices, mark sent/paid, resend PDF
admin/site-content-editor/	NEW	Edit hero title, subtitle, hero image, about section, social links
admin/chat-management/	NEW	Monitor all client-employee conversations, join as admin
admin/reports/	NEW	Date-range filters, CSV/PDF export, revenue by category, employee KPIs
client/chat/	NEW	Client chats with assigned employee; floating chat widget
client/invoices/	NEW	List and download payment invoices as PDF
client/file-delivery/	NEW	Download completed deliverable files uploaded by employee
employee/kanban/	NEW	Drag-and-drop Kanban board using Angular CDK (already installed)
employee/chat/	NEW	Employee responds to client messages per request
employee/file-upload/	NEW	Upload deliverable files for a completed request
core/services/theme.service.ts	NEW	On app init: GET /theme-settings/1 apply CSS custom properties to :root
core/services/chat.service.ts	NEW	Socket.io-client wrapper; connect/disconnect/send/receive methods
core/services/invoice.service.ts	NEW	Generate PDF from payment data using jsPDF; save to /invoices
core/services/file.service.ts	NEW	FormData upload to Express /uploads endpoint; returns Cloudinary/S3 URL
core/interceptors/retry.interceptor.ts	NEW	Auto-retry failed HTTP requests up to 3 times with exponential backoff

Path	Status	Description
shared/components/chat-widget/	NEW	Floating bottom-right chat bubble visible after login; expands to chat panel
shared/components/invoice-viewer/	NEW	Modal preview of invoice PDF before download
shared/components/file-uploader/	NEW	Drag-and-drop file input with progress bar and file type validation
shared/components/color-picker/	NEW	HSL color picker component used by theme-builder
core/models/invoice.ts	NEW	Invoice interface: id, paymentId, clientId, items[], total, pdfUrl, status
core/models/chat-message.ts	NEW	Message: id, requestId, senderId, senderRole, message, attachmentUrl, createdAt, isRead
core/models/theme-settings.ts	NEW	Theme: id, primaryColor, accentColor, logoUrl, fontFamily, brandName, darkMode
core/models/site-content.ts	NEW	Content: id (section key), heroTitle, heroSubtitle, heroImage, aboutText, socialLinks
admin/dashboard/	existing	Already exists — extend with real-time Socket.io data feed
admin/service-management/	existing	Already exists — extend with image upload + rich description editor
client/payments/	existing	Already exists — add SSLCommerz redirect button + invoice download
shared/components/notification-bell/	existing	Already exists — upgrade polling to WebSocket (Socket.io) real-time